

RSR Week 10 • Logistic Regression

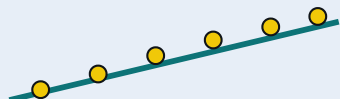
Blasket question set • 8 questions • thegraphcourses.org

Name _____
Date _____
Class _____

Conceptual quiz. Any example data or model output is introduced inside the question.

1 Which outcome is best modeled with logistic regression?

Continuous



e.g. blood pressure (mmHg)

Binary (yes/no)

0

1

e.g. disease: no / yes

Count

0, 1, 2, 3...

e.g. clinic visits / year

- A Average resting heart rate after a 6-minute walk
- B Whether a patient is cured of a disease (yes or no)
- C Number of malaria cases reported in a district in one year

2 What does a logistic regression model predict?

- A The residual for each observation
- B The probability of the outcome
- C The sample size needed for 80% power

3 You want to model a binary outcome cured from a continuous predictor dose. Which R call is correct?

- A `lm(cured ~ dose, data = trial)`
- B `glm(cured ~ dose, family = binomial, data = trial)`
- C `glm(cured ~ dose, family = gaussian, data = trial)`

4 A logistic model for weak grip from age was fit as:

$$\ln(p / (1 - p)) = -8.82 + 0.117 \times \text{age}$$

$$\beta_0 = -8.82 \quad \beta_1 = 0.117$$

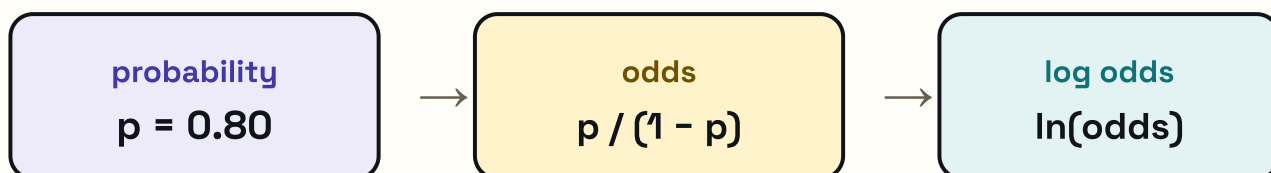
What does the coefficient 0.117 represent?

- A The change in the probability p for each extra year of age
- B The change in the log odds for each extra year of age
- C The residual standard error of the model

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Questions 5-8 • odds, odds ratios, and reporting language

5 If the predicted probability of an outcome is $p = 0.80$, what are the odds of that outcome?



A 0.20

B 0.80

C 4.0

6 A logistic model predicts weak grip from age. Part of summary (mod) is shown below. What is the odds ratio for a 1-year increase in age, and how do you get it?

```
# glm(weak_grip ~ age, family = binomial, data = klosa)
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -8.8243     0.3193   -27.64  <2e-16 ***
age           0.1168     0.0048    24.57  <2e-16 ***
```

A $OR \approx 0.117$; report the coefficient as-is

B $OR \approx 1.12$; exponentiate the age coefficient ($e^{0.1168}$)

C $OR \approx 0.50$; take 1 minus the age coefficient

7 An odds ratio of 0.75 for a binary exposure (exposed vs not) is best described as:

A 25% higher odds of the outcome among the exposed

B 25% lower odds of the outcome among the exposed

C The probability falls by exactly 0.75 percentage points

8 A paper reports: "Each extra year of age was associated with 12% higher odds of the outcome ($OR = 1.12$).” Which statement is a misconception?

A Each extra year multiplies the odds by about 1.12

B The probability of the outcome rises by 12 percentage points each year

C An OR of 1.12 is the same idea as adding a fixed amount on the log-odds scale

Answer Key • Week 10

Blooket / classroom key • cursor-grok-4.5-k7m2

Correct option is marked first. Keep this page out of the student Blooket set.

1B Binary yes/no outcome (cured or not).
Continuous and count outcomes need other models.

3B `glm(..., family = binomial)`
`lm / gaussian` is for continuous outcomes.

5C Odds = $0.80 / 0.20 = 4.0$
Odds = $p / (1 - p)$

7B OR 0.75 → 25% lower odds among the exposed.
 $1 - 0.75 = 0.25$

2B Logistic regression predicts the probability of the outcome.

4B 0.117 is the change in log odds per extra year of age.
 $\ln(p/(1-p)) = -8.82 + 0.117 \times \text{age}$

6B OR = $e^{0.1168} \approx 1.12$
Positive age coefficient → OR > 1

8B Misconception: OR 1.12 ≠ +12 percentage points in probability.
It is a 12% change in odds, not probability.